

Journal of Applied Statistics

ISSN: 0266-4763 (Print) 1360-0532 (Online) Journal homepage: www.tandfonline.com/journals/cjas20

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To cite this article: V. Gómez-Rubio, J. Lagos & F. Palmí-Perales (2025) Spatial similarity index for scouting in football, Journal of Applied Statistics, 52:14, 2745-2758, DOI: 10.1080/02664763.2025.2473542

To link to this article: https://doi.org/10.1080/02664763.2025.2473542

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JOURNAL OF APPLIED STATISTICS
2025, VOL. 52, NO. 14, 2745–2758
https://doi.org/10.1080/02664763.2025.2473542

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APPLICATION NOTE

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Spatial similarity index for scouting in football

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ABSTRACT

Finding players with similar profiles is an important problem in sports such as football (also known as soccer in some countries). Scouting for new players requires a wealth of information about the available players so that similar profiles to that of a target player can be identified. However, information about the position of the players in the field is seldom employed. For this reason, a novel approach based on spatial data analysis is introduced to produce a spatial similarity index that can help to identify similar players. The use of this new spatial similarity index is illustrated with an example from the Spanish competition 'La Liga', season 2019–2020, in which hundreds of players are clustered according to their position in the field.

ARTICLE HISTORY
Received 3 January 2025
Accepted 23 February 2025

KEYWORDS
Clustering; scouting; soccer; spatial analysis

1. Introduction

Scouting in sports refers to looking for players with a particular profile [11]. It can be conducted to fill a particular position within the team or to replace another player that might be injured or leaving the team. Hence, scouting plays an important role in sports management as well.

Typically, scouting can be performed by outlining the required skills, quantifying them and looking for players with similar values of these skills. Nowadays, different sports databases and websites will provide accurate statistics about different players that can be used for scouting [14].

Spatial data about the players is currently available on a regular basis and it is often used to produce heatmaps to show the players' positions during a game. This is often closely related to the role of the player within the team. However, this spatial information is seldom used to identify similar profiles of players according to their positions in the field.

Football data is usually obtained by either eventing or tracking [21,22]. Eventing refers to recording information about any relevant event (hence the name) during the game. For example, when a foul is committed, which is a relevant event during a game, variables recorded will include the player that produces the foul, player that receives the foul, position in the field, etc. Alternatively, tracking refers to recording information using different

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Supplemental data for this article can be accessed online at http://dx.doi.org/10.1080/02664763.2025.2473542

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devices (including TV broadcast). For example, the position of a player may be recorded using a small GPS so that other variables such as distance covered during a game, etc. can be computed. In general, data collected in eventing will require more human intervention than the one obtained with tracking.

Recent reviews on scouting [34,35] have surveyed the current state of research in this field, discussing key findings and developments. In addition, other authors [15] proposed several promising yet unexplored research directions. The primary focus of this study, however, is on quantitative analysis rather than qualitative evaluations, which have also been addressed in prior research [27].

In terms of quantitative analysis, a number of studies employ machine learning techniques to derive insights from extensive datasets. For instance, Pavitt *et al.* [23] utilises different methods to develop an application that compiles statistics for individual players and leverages machine learning and rule-based models to examine player relationships and statistical interdependencies. Similarly, Vilela *et al.* [32] employs data mining and predictive modelling to evaluate player performance using a variety of machine learning strategies available through the rminer package [5].

Machine learning is also applied in additional studies such as [7], where a random forest algorithm is used to simulate and assess a new player's fit within a team. This approach essentially creates a variable representing the player's 'fit' within the team, offering a means of quantifying the player's 'value' through a single composite indicator. This approach aligns with other quantitative studies in scouting that focus on developing composite indicators for ranking players, such as [25,26].

Other quantitative methodologies in scouting include the use of multiple regression models to analyse relationships between specific scouting variables [28]. However, more complex models that analyse players' performance through position-tracking data are relatively uncommon. For example, Kempe *et al.* [13] models pass effectiveness by analysing player coordinates on the field. The study develops a composite indicator for passes and fits a ridge regression model to explain variability in this indicator, observing associations between pass location, speed, and angle, though the model's overall fit was limited.

As seen in the aforementioned study, spatial modelling has been explored by various researchers in football. Some studies focus on modelling player and ball positions using motion models [9,20]. Other studies have employed alternative methods to capture spatial and temporal aspects of field dynamics without employing a fully spatio-temporal model. For instance, passing trends have been modelled using motion models [29] or basic linear models such as ANOVA [31]. Furthermore, position-tracking data has been used to analyse player or team positions through descriptive statistics and the development of various metrics [8,17].

While there are multiple approaches to incorporating spatial coordinates into the modelling process, many studies do not employ formal spatial models or spatial techniques. Instead, they tend to model coordinates or centroids using alternative methods. However, a few studies propose models within a spatial framework. For example, Yeung *et al.* [33] presents a model for the timing, location, and types of in-game football events based on a Transformer-Based Neural Marked Spatio-Temporal Point Process (NTPP), which combines point process modelling with machine learning techniques.

Alternatively, Narayanan *et al.* [19] proposes a Bayesian framework for inference, offering a comprehensive model of flexible marked point processes suitable for analysing

sequences of in-game events in football. This model allows for the simulation of in-game events and estimation of event probabilities, enabling calculations of expected goal value at specific moments within a game. Despite these contributions, spatial methodologies in football analysis remain scarce, with few studies offering formal spatial modelling approaches.

Given the lack of methods that consider spatial data in football for scouting, a spatial similarity index is proposed to address this issue by exploiting the spatial information available about the players. In particular, the spatial information will be represented using a lattice of the field so that it is divided into small squares or rectangles for which a measure of the occupation or activity of the player is available (e.g. the average time spent during a game). Then, these values will be compared using appropriate spatial cross-correlation statistics such as Lee's statistic [16,18], from which a spatial similarity index is derived. Although the computation of the spatial similarity index will be illustrated with an example from football [2], it is suitable for similar sports.

This paper is organised as follows. Section 2 provides an introduction to spatial data analysis. Methods to assess spatial similarity are discussed in Section 3. Next, an example is developed in Section 4. Finally, a discussion and some concluding remarks are provided in Section 5.

2. Spatial data analysis

Spatial data analysis refers to a particular area of statistics that deals with data for which a location is available. Several authors [4,10] provide a detailed summary of the different types of data and the corresponding statistical methodology to analyse them.

In this particular case, the spatial domain will be the field and it will be assumed that it has been divided into smaller regular areas (usually squares or rectangles), so that a number of relevant variables have been recorded for each area. Typically, this will be the average number of minutes spent in each area or, alternatively, the proportion of time spent in each area.

Figure 1 illustrates how a football field can be split into smaller regular areas (left plot), using the R package ggsoccer [30]. Note that each area is represented by a polygon and there are a number of variables that could be recorded. For example, the number of yellow or red cards issued by an action in each area, average number of players in each area and many others. For the current analysis, the focus will be in the time spent by each player in each area and developing a suitable methodology to assess their similarities. Furthermore, Figure 1 (right plot) includes a representation of the adjacency between the different small areas. Blue lines link nearby areas that will be designated as neighbours, i.e. squares that share at least one point of the boundary will be considered as neighbours when computing the spatial similarity index described below.

Figure 2 shows heatmaps for 5 different players. From a statistical point of view, this representation can be regarded as a summary of the available data. From a scouting perspective, given a target player with a particular spatial profile the aim is to find another player with a very similar spatial profile.

When dealing with spatial data, the variables represented will often show some spatial pattern, pointing to the fact that values from neighbouring regions are not independent.

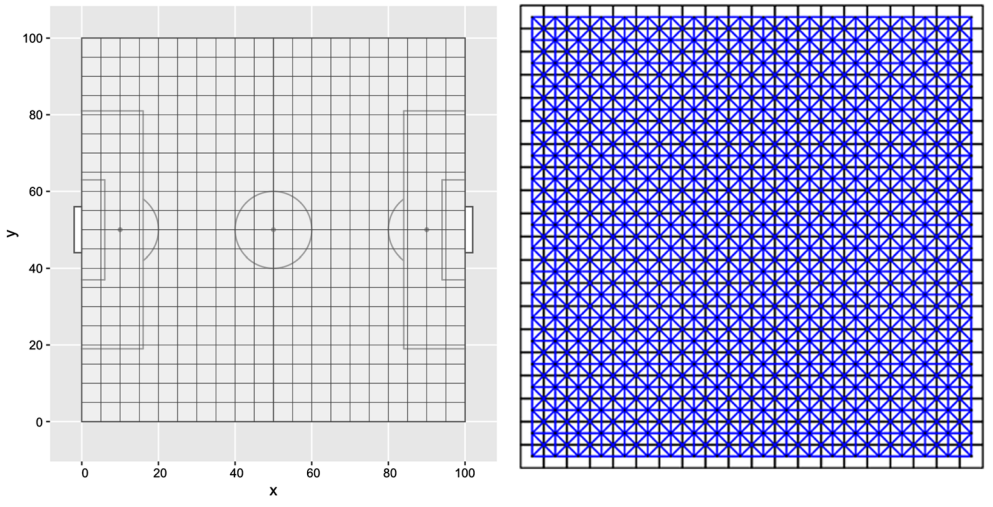


Figure 1. Spatial representation of a football field using small areas (left) and representation of the adjacency (blue lines) between the different small areas (right).

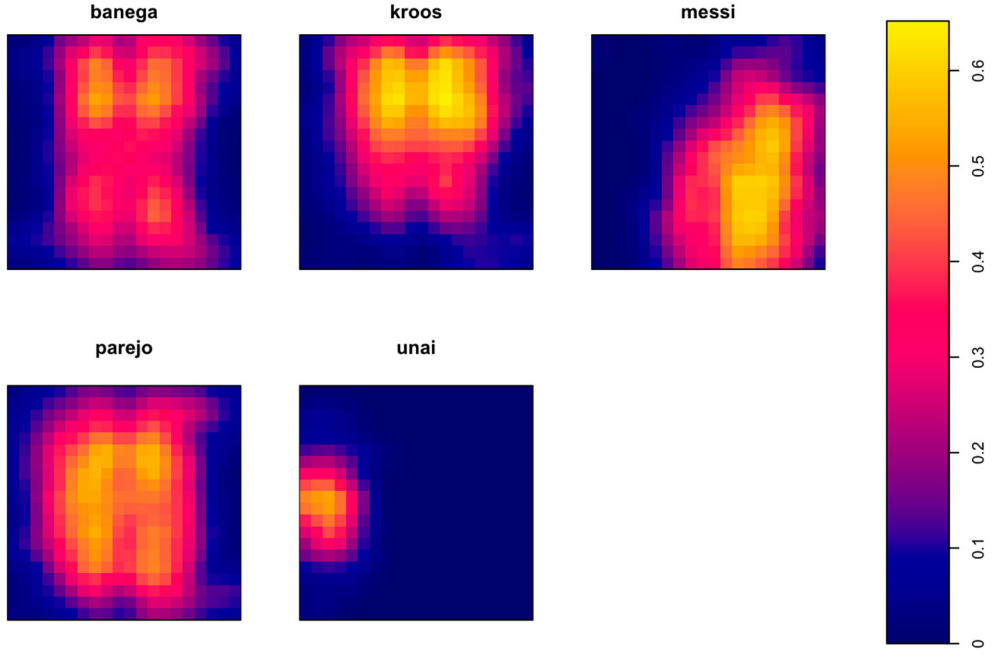


Figure 2. Spatial distribution of 5 different players from 'La Liga', season 2019–2020.

This means that values from areas that share a boundary (i.e. neighbours) will have similar values. This spatial autocorrelation can be tested in different ways, as explained in Section 2.1.

Regarding neighbourhood, it can be represented as a binary matrix with as many rows and columns as the number of areas so that entries with a value of 1 mean that areas *i* and

j are neighbours and an entry of 0 means that they are not. In addition to neighbourhood, spatial weights can be defined between any pair of areas to measure the strength of their relationship. However, most often spatial weights will be taken as 1 and 0, in a similar way as neighbourhood. However, other alternatives are available [4]. The matrix of spatial weights will be represented by W.

2.1. Spatial autocorrelation

A convenient way to test for spatial autocorrelation is Moran's I statistic [4]. It is defined as

$$I = \frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Here, n refers to the number of areas, w_{ij} is the spatial weight between areas i and j, x_i is the value of the variable of interest in area i and $\bar{x}$ the mean of $\{x_i\}_{i=1}^{n}$.

Moran's I will be positive when either areas with values over the mean $\bar{x}$ tend to have neighbours with also values over the mean or when areas with values below the mean tend to have neighbours with values also below the mean. When the opposite happens, the value of Moran's I will be negative. When there is no spatial structure in the data, the value of Moran's I will be close to zero.

Hence, positive values of Moran's I indicate positive spatial association. A statistical test can be developed to assess whether the value of Moran's I statistic is significantly high [4].

It is important to note that the presence of spatial autocorrelation will depend on the nature of the phenomenon under study. In our particular case, it is reasonable to find spatial autocorrelation because players move smoothly throughout the field and tend to occupy different parts of the field depending on their role in the team.

2.2. Spatial cross-correlation

Moran's I statistic has been introduced in order to outline the important elements to consider when assessing spatial autocorrelation. However, Moran's I cannot deal with more than one variable. In order to compare the spatial distribution of two variables, Lee's statistic [16] will be used.

Given a set of n areas with associates weights $\{w_{ij}\}_{i,j=1}^{n}$, where variables $\{x_i\}_{i=1}^{n}$ and $\{y_i\}_{i=1}^{n}$ have been measured, Lee's statistic is defined as follows:

$$L = \frac{n}{\sum_{i=1}^{n}(\sum_{j=1}^{n} w_{ij})^2} \frac{\sum_{i=1}^{n}\left(\sum_{j=1}^{n} w_{ij}(x_i - \bar{x})\right)\left(\sum_{j=1}^{n} w_{ij}(y_j - \bar{y})\right)}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Note that Lee's statistic will have positive values when both variables tend to have values above or below their respective mean values, similarly as it happened with Moran's I statistic. Lee's statistic will provide negative values when one variable tends to have values above the mean and the other variable tends to have values below its mean.

Furthermore, a statistical test using the expectation and variance of the statistic under spatial independence can be conducted. Note that the alternative hypothesis of interest is that of positive spatial correlation as this implies that two players tend to occupy similar

areas in the field. A negative spatial association (which will yield a negative value of Lee's statistic) is not of interest because this implies that two players will tend to occupy different positions in the field.

3. Spatial similarity

In order to provide a measure of spatial similarity, Lee's statistic will be used to assess spatial association between the spatial distribution of each pair of players. Note that this statistic can be computed on a number of variables, e.g. average time spent or proportion of time in each area, and that this will lead to values of the test statistic in different scales. Note also that different players may be more prone to play and, hence, their values may be higher, leading also to a changes of scale when computing Lee's statistic. For this reason, it is advisable to compute Lee's statistic using the proportion of time spent in each square of the field.

As a metric to measure spatial similarity between two players, the standardised Lee's statistic (i.e. L minus its expectation divided by the standard deviation) or the associated p-value of a one-sided test (see Section 2.2) could be used. However, to have a bounded measure of association that is independent from the scale, the aforementioned p-value will be used. In this way, values close to zero will indicate a strong spatial association while values close to 1 will mean that there is either no spatial association or this is negative. In any of these cases, the two players will not be similar in terms of their spatial position in the field.

To sum up, the associated p-value can be regarded as a *pseudo-distance* in the space of players. It is worth noting that this is not a distance in the strict mathematical sense but a pseudo-measure of dissimilarity, i.e. the higher the value the less similar the players are. If a similarity measure is needed, then one minus the p-value can be taken.

Once the similarity measure has been obtained, players could be grouped together by means of different clustering algorithms so that a hierarchical clustering can be conducted [6]. This hierarchical clustering will be useful in order to group players together. Other clustering methods could be employed together with the pseudo-distances in order to group the players as well.

4. Example: spatial distribution of soccer players

Two examples will be developed in this Section to illustrate the methods described in previous sections. First of all, an example with only 5 players (from Spanish competition 'La Liga', season 2019–2020) is included in Section 4.1 to provide some insight on how the proposed methodology works. Next, an example with hundreds of players is discussed in Section 4.2 to illustrate a real application of the proposed methodology. Note that due to confidentiality reasons only the datasets for the toy examples are available.

Data for the examples below have been acquired from data provider Wyscout (https://www.hudl.com/en_gb/products/wyscout). This is a subset of a major dataset that corresponds to players from different leagues worldwide, season 2019–2020. Altogether, the original dataset comprises 14914 players from 1389 teams worldwide. However, for the purpose of this paper, only the 500 players from 20 teams in the Spanish football competition 'La Liga', season 2019–2020, are considered.

Data comes as a set of coordinates that represent the centroids of some squares in the field after transforming the game field into a square in $[0, 100] \times [0, 100]$. Attached to each coordinate is a value that represents the activity of the player in that area of the field (e.g. average time spent in that square).

In order to match the data from the different players to the same regular grid (see Figure 1, left plot) to compute the similarity measure, a weighted density smoothing will be conducted using the centroids and the associated value as weights so that a raster image is created for all players. This will ensure that all players' data are represented using the same raster image so that the regular lattice and its adjacency structure can be computed.

The analysis has been conducted on a Linux cluster running Ubuntu with 64 cores Intel(R) Xeon(R) CPU E5-2683 v4 @ 2.10GHz and 157 Gb of RAM. The main R packages used in the analysis have been spatstat [1] and sp [3] for spatial data handling and visualisation and spdep [24] to compute Lee's statistic and the associated spatial test and p-value. R package `factoextra` [12] has been used to visualise the distances between the players.

4.1. Toy example

For this first example, data about 5 different players from the Spanish competition 'La Liga' (season 2019–2020) will be used. These players are Unai Simón (goalkeeper, Atlético de Bilbao), Éver Banega (midfilder, Sevilla F. C.), Toni Kroos (midfilder, Real Madrid), Dani Parejo (midfilder, Valencia F. C.), and Lionel Messi (forward, F. C. Barcelona). This selection of players covers different roles and distributions on the field. These players have been chosen to illustrate the proposed methodology because of their distinctive spatial patterns that cover different positions in the field (goalkeeper, midfilder and forward).

Figure 2 shows the different spatial distributions of the players. As it can be seen, there are clear differences between some of them while there are also clear similarities among the midfilders.

In order to provide some insight on the proposed methodology to identify similar players, Table 1 shows the values of Lee's statistic when comparing each pair of players (diagonal and lower triangle) as well as the associated p-value of the one-sided test (upper triangle). Note how the values of Lee's statistic when a player is compared to himself are quite large, but note also that these are different from one player to another. These values lead to very small p-values which, in practice, can be taken as zero.

Regarding the comparisons between two different players, Unai has a negative Lee's statistic when tested against all the other players given his distinctive spatial distribution in the field. In particular, all values of Lee's statistic are negative when compared to other players as field occupancy is essentially different to all the other players compared. This means a negative spatial association and a very small p-value.

When comparing all the other players, Lee's statistic is always positive and p-values are very small. However, by looking at the values of the statistic, it is possible to observe that all three midfilders (i.e. Banega, Kroos and Parejo) have larger values among them than when compared with Messi (who is a forward).

Next, a hierarchical clustering will be conducted in order to determine the different groups of players in the dataset. The algorithm starts with each player being a different cluster and then clusters are aggregated in turn so that the two most similar clusters are

Table 1. Values of Lee's statistic (diagonal and lower triangle) and associated *p*-value from one-sided test (upper triangle) for 5 different players from 'La Liga', season 2019–2020.

		Player				
		Banega	Kroos	Messi	Parejo	Unai
Player	Banega	49.92	0	0	0	1
	Kroos	42.78	53.98	0	0	1
	Messi	15.42	3.77	56.49	0	1
	Parejo	42.05	36.01	15.87	49.70	1
	Unai	−22.72	−15.72	−27.52	−7.81	52.13

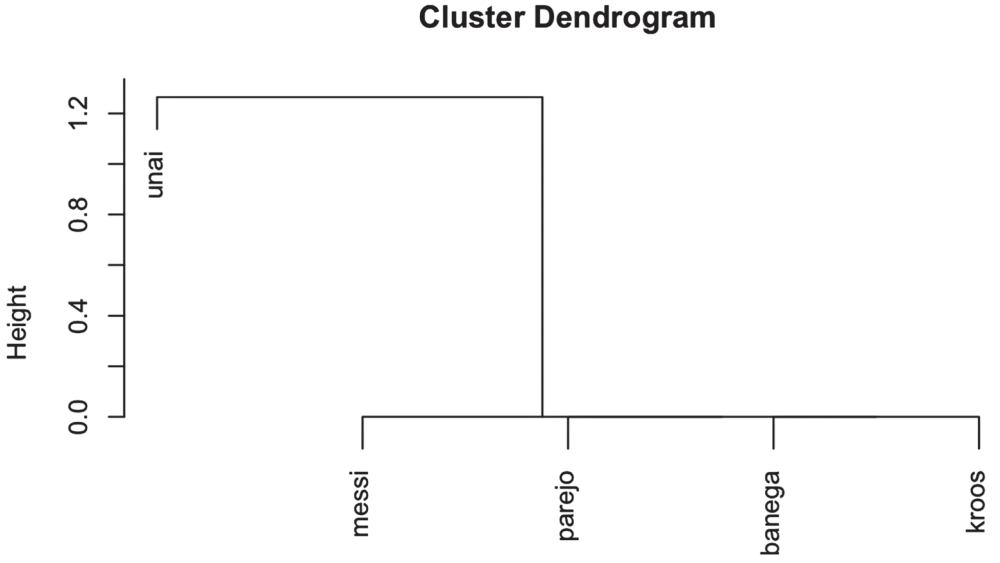


Figure 3. Dendrogram of a hierarchical clustering for 5 different players from 'La Liga', season 2019–2020.

combined in each step until all players are together in one cluster. The distance between two clusters is the largest distance between any player in the first cluster and any player in the second one. This is implemented in the R function `hclust()` when the `method` argument is set to `'complete'` (the default). Note that considering other distances between the cluster may lead to slightly different clusterings. The manual page of function `hclust()` provides detailed information and references about the clustering methods implemented.

Figure 3 shows the dendrogram from the hierarchical clustering using the pseudo-distance computed using the *p*-value from Lee's statistic. As it can be seen, the hierarchical clustering seems to identify two different groups according to the different roles of the players in the example.

Note also that the pseudo-distances take very extreme values. On the one hand, when Unai is compared to all the other players, the values are very close to zero. On the other hand, when all players but Unai are compared, the values of the pseudo-distances are very close to 1. This makes the dendrogram difficult to interpret as all pseudo-distances are, in

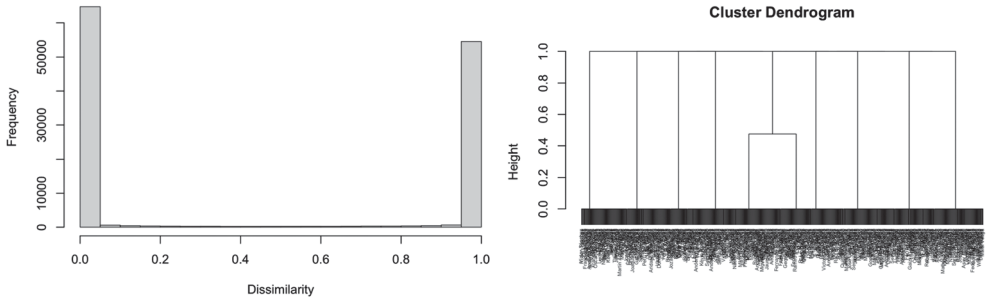


Figure 4. Distribution of spatial similarity index values (left) and resulting dendrogram of a hierarchical clustering (right) for all player considered from 'La Liga', season 2019–2020.

practice, either zero or one. In fact, they are not exactly zero or one but the tiny differences cannot be appreciated in Table 1.

However, by inspecting how the clustering is done, more insight on how the pseudo-distance are used can be gained. First, Banega and Kroos are put together in the first iteration. This is not surprising as Banega and Kroos have the highest value of Lee's statistic as well as the smallest p-value. In successive iterations, Parejo, Messi and Unai are added (in this order) to the previous cluster. Again, Parejo is added first because he is the most similar player to Banega and Kroos as he is also a midfielder. He is followed then by Messi (a forward) and, finally, by Unai (a goalkeeper).

Although this is a small example, it has illustrated how the proposed methodology to compare and cluster players according to their spatial distribution in the field works. Next, a more realistic example comprising a larger number of players and teams is developed.

4.2. Analysis of 'La Liga', season 2019–2020

In order to illustrate the use of the spatial similarity index, it will be applied to the players in the Spanish competition 'La Liga', season 2019–2020, which is the top level football league in Spain. This requires comparing and computing the index for each pair of players. However, this is something that can be easily done in parallel and the computations of the index are simple.

Furthermore, not all players need to be compared at the same time and they can be grouped according to their role. For example, goalkeepers do not need to be compared with any other players. Also, different variables can be considered prior to computing the spatial similarity index. However, to best illustrate the performance of the proposed methodology, all players will be compared to each other.

Hence, the similarity index has been computed using the methodology described in Section 3 and a hierarchical clustering computed. The left plot in Figure 4 summarises the values of the pseudo-distances obtained when comparing all the players. As it can be seen, most of the values are very close to either 0 or 1, with just a few in between. This points to the fact that Lee's statistic is quite strict about detecting similar patterns.

As stated above, a hierarchical clustering has been conducted on the dataset and it can be seen in the right plot in Figure 4. The dendrogram seems to define 10 main groups, with different number of players in each of them. Furthermore, setting a cut-off point of 0.001

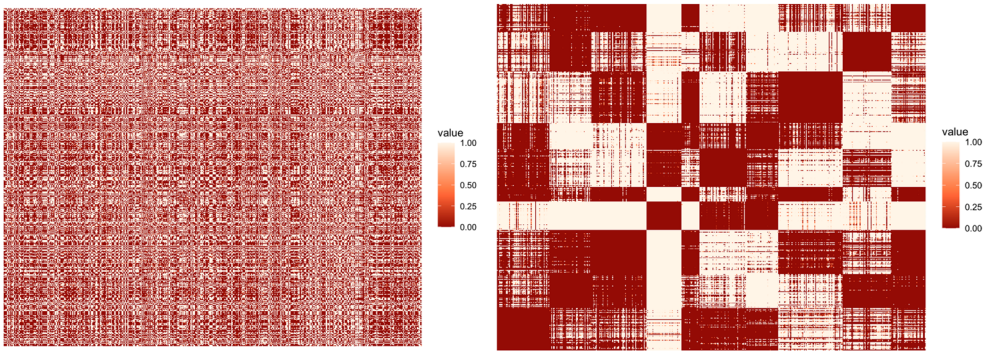


Figure 5. Matrix of similarities between players ordered by team (left) and after reordering according to the clusters found (right).

seems reasonable in order to break the players down into groups. As a result, 10 groups of players are obtained, with an average number of players of 50 and a range between 21 and 75.

Similarly, Figure 5 shows matrix of the distances between each pair of players, so that each row and column represent a player. Darker colours indicate a shorter distance (i.e. similar position in the field) while lighter colours mean a longer distance (i.e. different position in the field). In the left plot, players are sorted by team (the order in the original data). Hence, no positional clusters can be identified. In the right plot, players have been reordered according to the 10 clusters found in the data. For this reason, 10 blocks can be easily identified in the minor diagonal (from down-left to top-right). The plots have been produced with R package `factoextra` [12].

Regarding the clusters, they seem to identify clearly the different types of players in the data. The heatmaps of the players in the different clusters have been included as Supplementary Material. However, in Figure 6 the heatmaps of players in cluster 4 have been included. As it can be seen, this cluster clearly identifies the goalkeepers in the dataset. For these reasons, it is possible to conclude that the similarity index is a suitable metric to group players according to their position in the field during the game.

5. Discussion and concluding remarks

As it could not be otherwise, the role of a player in the team is intimately linked to its position in the field. Hence, by exploring the spatial distribution it is possible to obtain a sensible metric for comparing different players. Players with similar spatial distributions will, most likely, have similar roles within their respective teams. Hence, by comparing these spatial distributions to find similarities and dissimilarities it will be possible to match similar players. This is particularly important to replace injured players or to look for similar players in the market.

The methodology described in this paper can effectively discriminate among a large number of players according to their spatial distribution in the field during the game. Note that this spatial similarity index can be used after players have been filtered according

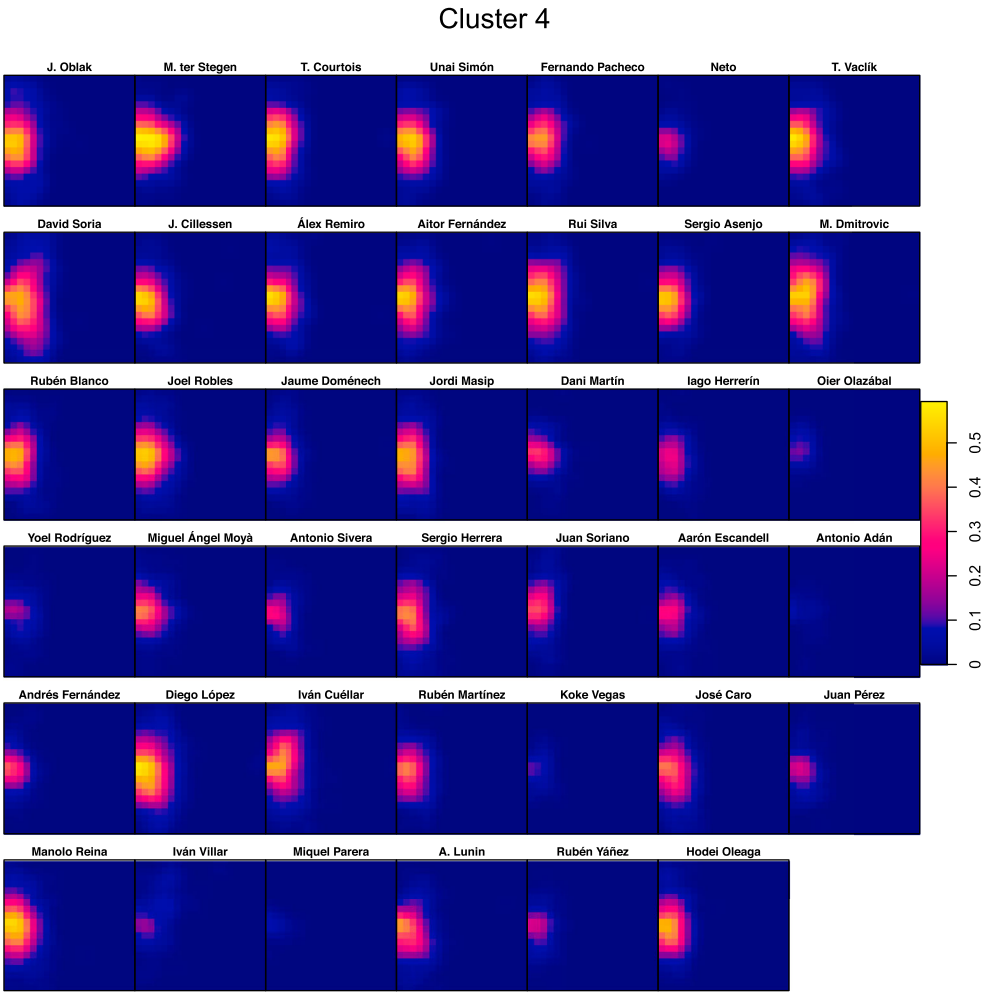


Figure 6. Heatmaps of players in cluster 4, which clearly identifies the goalkeepers.

to other characteristics. Hence, by combining this spatial similarity index with other attributes, a more effective scouting can be achieved.

Note that the spatial variable measured can be the (average) time spent in each area of the field but it can also be tailored in different ways to match specific needs. For example, specific spatial profiles can be produced for each player when the team is attacking or defending, so that the target player should now match these profiles rather than a single profile. This means that the location of a player is influenced by the team's tactics, which may lead to different spatial profiles according to how the opposing team is organised as well.

Similarly, when tracking data is used, the spatial profiles can be used to inspect the position of the players that do not take part in the main action with the ball. This can lead to off-ball movements that may be important during the game. For example, a midfilder may occupy parts of the field near the area where a potential pass can be received so that it takes defenders away from where the main action is taking place.

Although the example developed in this paper describes an application to football, the spatial similarity index and the methodology presented in this paper could be used for other similar team sports. Furthermore, the spatial similarity index could be combined with other existing data to better identify similar profiles.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work has been supported by grant PID2022-136455NB-I00, funded by MCIN/AEI/10.13039/501100011033/FEDER and the European Regional Development Fund, grant CIAICO/2022/165, funded by Dirección General de Ciencia e Investigación (Generalitat Valenciana, Spain), and grant SBPLY/21/180501/000241, funded by Consejería de Educación, Cultura y Deportes (Junta de Comunidades de Castilla-La Mancha, Spain) and FEDER.

Code availability

Code and datasets to replicate the results in this paper are available at https://github.com/becariopre cario/spatialindex. Note that raw data are not provided due to being commercial data but the heatmaps used to compute Lee's statistic are provided.

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